

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. in Artificial Intelligence & Data Science

Semester : VII

Subject Code & Name: BTAIC703 Data Engineering

Max Marks: 60

Date:06-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define data engineering. With suitable diagram, explain data engineering process.	Understand/CO 1	6
B) Define data pipeline and differentiate between ETL and ELT pipelines.	Understand/CO 1	6
C) Explain the importance of EDA in data processing.	Understand/CO 1	6
Q.2 Solve Any Two of the following.		12
A) With examples, differentiate between following functions in Microsoft Excel, i. sumif() and sumifs() ii. averageif() and averageifs() iii. countif() and countifs()	Understand/CO 2	6
B) With suitable example, explain vlookup() used for finding duplicates from two columns in Excel.	Apply /CO 2	6
C) Demonstrate pivot table geography by considering any Sales data	Understand/CO 2	6
Q.3 Solve Any Two of the following		12
A) With suitable diagram, describe Power BI architecture.	Understand/CO 3	6
B) Consider any Sales data and explain concept of calculated column and calculated measure in Power BI.	Apply/CO 3	6
C) Explain different types of filters used in Power BI.	Understand/CO 3	
Q. 4 Solve Any Two of the following.		12
A) What are different tableau products? Also describe concept of dimensions and measures in Tableau.	Understand/CO 4	6

B)	How to create donut chart and heatmap in Tableau? Explain the process with suitable example.	Apply/CO 4	6
C)	What is data modelling? Explain different types of joins in Tableau.	Understand/CO 4	6
Q.5	Solve Any Two of the following.		12
A)	With suitable diagram, explain the components of an automated data pipeline. Enlist the steps for creating automated data pipeline.	Understand/CO 5	6
B)	How do you record a macro in Excel? Write a simple VBA macro to add two numbers in Excel.	Apply/CO 5	6
C)	Describe any six possible ways for sharing and collaborating your data in Power BI.	Understand/CO 5	6

***** End *****